Mohamed Khaled Mahmoud

Backend / Full Stack .NET Developer

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Profile

Backend .NET Developer working on production systems at Appssquare: owner of 4 modules in a .NET 10 modular monolith (DDD, CQRS/MediatR, EF Core) and contributor to a live .NET 9 marketplace. Focused on correctness, backing changes with automated tests and verifying releases against real logs. ITI Full Stack .NET graduate with Angular and AI-integration experience (Semantic Kernel, RAG); B.Sc. in Computer and Control Engineering with honors, ranked 2nd in class.

Technical Skills

- **Back-End:** C#, .NET 10/9, ASP.NET Core (Web API, MVC), Entity Framework Core, LINQ, RESTful APIs, JWT with refresh-token rotation, ASP.NET Identity, FluentValidation, SignalR, Serilog, OpenAPI (Swagger, Scalar)
- **Architecture:** Clean Architecture, Modular Monolith, Domain-Driven Design, CQRS, MediatR, SOLID Principles, Repository & Unit of Work, Design Patterns, OOP
- **Front-End:** Angular (Standalone Components, Signals, RxJS, Reactive Forms, Lazy Loading), TypeScript, JavaScript (ES6+), HTML5, CSS3, Bootstrap
- **Databases:** Microsoft SQL Server, T-SQL, Query Optimization & Indexing, Redis (Caching)
- **Testing:** xUnit, NUnit, Moq, Integration Testing (WebApplicationFactory), Contract Tests, Architecture Tests
- **DevOps & Cloud:** Docker, Docker Compose, CI/CD with GitHub Actions, Microsoft Azure (App Service, Azure SQL, Blob Storage), Linux (systemd, Kestrel, Apache reverse proxy), Git (GitHub, GitLab)
- **AI & Automation:** Semantic Kernel, RAG (Retrieval-Augmented Generation), Qdrant vector search, Prompt Engineering, n8n
- **Languages:** Arabic (Native), English (Professional Working Proficiency)

Experience & Training

Appssquare Hybrid
Backend .NET Developer (Intern) *Sep 2026 – Present*

Kama Kan, Real-Estate Inspection Platform *(.NET 10, in development)*

- Own 4 of 7 modules (Identity, Properties, Rental Contracts, Payments) in a modular monolith built with DDD (aggregates, value objects, strongly-typed IDs), CQRS/MediatR and EF Core on SQL Server; module boundaries are enforced by build-breaking architecture tests.
- Built the Identity module: phone OTP sign-up, JWT with refresh-token rotation and reuse detection, role policies checked against live account status, and rate limiting per IP, phone and account.
- Reviewed the payments module and uncovered 3 proven bugs and 8 untested money rules; fixed the 3 bugs with regression tests, including a SQL Server race condition on the payment deadline.
- Contribute to a 1,278-test suite (xUnit domain, database-backed, contract and HTTP integration tests); fixed a thread-safety bug behind intermittent test failures.

Khayat, Tailoring Marketplace *(.NET 9, live in production)*

- Shipped an admin account-lifecycle feature (activate, suspend, delete) with safety guards that block irreversible deletions while a user holds a wallet balance or open orders.
- Closed a production security gap by enforcing role-based authorization across admin endpoints, auditing live accounts first so no existing user was locked out.
- Diagnosed production incidents from structured Serilog logs, including a failure that blocked every sign-up and login whenever SMS OTP delivery failed; deploy releases to a Linux server (systemd, Kestrel behind Apache) and verify them through health checks and logs.

Information Technology Institute (ITI) – MCIT Tanta, Egypt
Full Stack Web Development using .NET, Intensive Training Program *Jan 2026 – Jun 2026*

- Led a 5-developer team on EcoBazar, a multi-role grocery e-commerce app in vanilla JavaScript (ES6 modules, OOP), running a Git feature-branch workflow with pull-request reviews.
- Built RESTful .NET back-ends with Clean Architecture and CQRS/MediatR, and Angular front-ends with standalone components, Signals and reusable UI modules; containerized services with Docker.
- Designed SQL Server schemas with Entity Framework Core, tuning LINQ queries and indexes for fast data retrieval.

Egyptian Armed Forces Egypt
Mandatory Military Service, Completed *Nov 2024 – Dec 2025*

– Continued self-directed study in .NET and Angular throughout service.

Outlier Remote
AI Model Trainer (Freelance) *May 2024 – Oct 2024*

– Evaluated generative AI outputs across 50+ review cycles against structured rubrics, and produced annotated training data through prompt engineering.

Information Technology Institute (ITI) – MCIT Mansoura, Egypt
Backend Web Development using .NET, Summer Code Camp *Jul 2023 – Aug 2023*

– Built RESTful APIs with ASP.NET Core and Entity Framework Core, reducing redundant queries through projection and eager loading, and validated endpoints with Postman.

Projects

TrackManagement, Music Distribution Platform | *.NET, EF Core, SQL Server, JWT, Angular* GitHub

– Applied DDD in a Clean Architecture solution (Domain, Application, Infrastructure, API): a track moves Draft → Submitted → Distributed only through guarded domain methods, and ISRC codes are validated as a value object.
– Scoped JWT authorization to the one endpoint with an external side effect (distribution) through a dedicated policy, with FluentValidation on all inputs.
– Built the Angular client with artist, genre and status filters and an auth interceptor; caught and fixed integration bugs (missing CORS policy, missing eager loading) by running the full stack end to end.

EstateFlow AI, Real Estate Platform | *.NET 10, Angular, Redis, Semantic Kernel* GitHub

– Architected the system as a modular monolith with CQRS/MediatR, keeping modules loosely coupled and independently deployable.
– Implemented Redis caching with write-invalidation on high-traffic listing endpoints, serving repeated queries from cache instead of the database.
– Built AI features with Semantic Kernel: a RAG assistant that answers property questions, Qdrant vector search that flags duplicate listings, and Azure Document Intelligence OCR that extracts lease start and end dates; an n8n workflow routes each maintenance ticket to the responsible party.
– Deployed to Azure (App Service + Azure SQL) through a GitHub Actions pipeline that builds and deploys on every push to main.

Vortex, Multi-Vendor E-commerce Platform | *ASP.NET Core, SignalR, Angular* GitHub

– Designed the core database schema and the REST API for the product catalog, cart and checkout pipeline using CQRS/MediatR.
– Built role-based Angular dashboards (Admin, Seller, Customer) with route guards, receiving real-time order and stock updates over SignalR.
– Integrated Stripe and PayPal payments with automated stock deduction and email notifications on order status changes.

AI-Powered Floor Plan Generator (B.Sc. Graduation Project) | *Python, Deep Learning* GitHub

– Generated 2D and 3D architectural floor plans from basic sketches with deep learning, served through REST APIs.

Education

Higher Institute of Engineering and Technology Kafr El-Sheikh, Egypt
B.Sc. in Computer and Control Engineering *Aug 2019 – Jul 2024*

– **Grade:** Excellent with Honors, **ranked 2nd** in the 2024 graduating class.